Enhancing Drug Demand Forecasting & Inventory Management Using Machine Learning

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Group Name: 3Mtt Group C2 Advanced Data Analysis (108)

Problem Statement

1. **Drug Stockouts**
   Pharmacies run out of high-demand drugs, leaving patients without critical medication.

2. **Expiry Waste**
   Overstocked drugs expire before being sold, causing significant financial losses and patient dissatisfaction.

3. **No Forecasting**
   Inventory decisions are made manually with no data-driven demand prediction system.

Why This Matters

- Patient safety is at risk when life-saving drugs are unavailable
- Pharmacies lose revenue when stock expires before sale
- Manual inventory systems cannot scale to hundreds of drugs
- A data-driven system can predict demand and prevent both problems simultaneously

SOLUTION

Project Overview & Solution

This project builds a machine learning system that analyses pharmacy transaction data to predict drug demand, flag stockout risks, and identify drugs about to expire thereby giving pharmacy managers actionable intelligence.

01

Demand Forecasting

To Predict how many units of each drug will be sold using a Random Forest model trained on historical transactions.

02

Stockout Alert System

Classify which drugs will run out of stock within 3 days based on current inventory and predicted sales rate.

03

Expiry Risk Detection

Flag drugs that are likely to expire before being sold, enabling timely discounting or redistribution.

04

Power BI Dashboard

Export clean, model-enriched data to Power BI for an interactive inventory management dashboard.

Tools & Technologies

Data Handling

pandas · NumPy

Loading, cleaning, feature engineering of the 700-row pharmacy dataset

Machine Learning

scikit-learn (Random Forest)

Training regression and classification models for demand and risk prediction

Visualisation

Matplotlib . Seaborn . Plotly

Exploratory Data Analysis

Dashboard

Microsoft Power BI

Interactive inventory management dashboard for pharmacy decision-makers

Environment/platform

Jupyter Notebook · Python 3

Step-by-step analysis workflow combining code, charts, and narrative

Export

CSV · Excel (.xlsx)

Clean data outputs for Power BI ingestion and stakeholder reporting

Methodology & Implementation

1 Data Loading & Inspection

700 pharmacy records loaded. Confirmed zero missing values. Converted dates, checked distributions.

2 Feature Engineering

Created days_to_expiry, stock-status, Year, Month, Days_to_Stockout, expiry risk flag, stockout risk flag etc.

3 Exploratory Data Analysis

Ranked demand by category, trended sales over time.

4 Model Training

Three models trained: Random Forest Regressor (demand), Random Forest Classifier (stockout risk), Random Forest Classifier (expiry risk).

5 Evaluation

Random Forest Regressor scored with MAE & RMSE. Random Forest Classifiers evaluated on Accuracy, Precision, and Recall.

6 Export & Dashboard

Encoded columns removed. Clean CSV exported. Loaded into Power BI for interactive visualisation.

Results & Model Performance

Here is what our models achieved — explained in plain terms:

Model 1 — Demand Forecasting (Random Forest Regressor)

13.10
MAE
Mean Absolute Error

15.38
RMSE
Root Mean Sq. Error

The Random Forest model achieved a MAE of 13.1 and a RMSE of 15.4, indicating a moderate level of prediction accuracy. Although the model's average deviation suggests some level of forecasting uncertainty, the predictions are sufficiently reliable to support inventory planning when combined with appropriate safety buffers.

Model 2 — Stockout Risk Classifier

91%
Accuracy
Stockout Risk

91%
Precision
Stockout Risk

91%
Recall
Stockout Risk

Model 3 — Expiry Risk Classifier

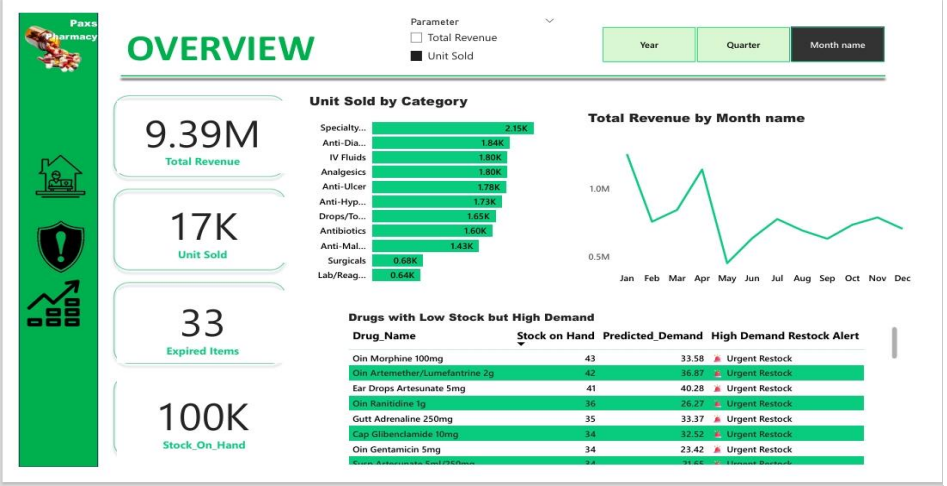
99%
Accuracy
Expiry Risk

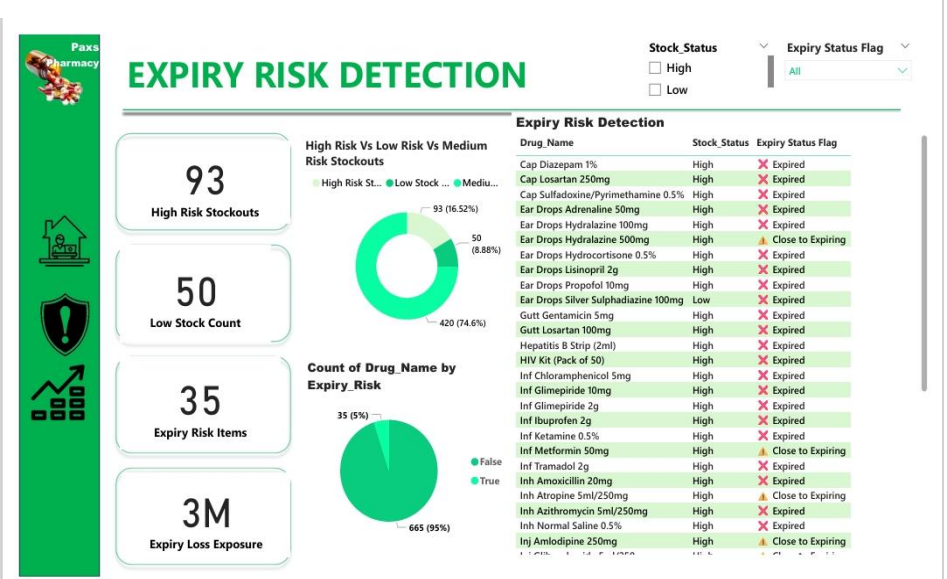
99%
Precision
Expiry Risk

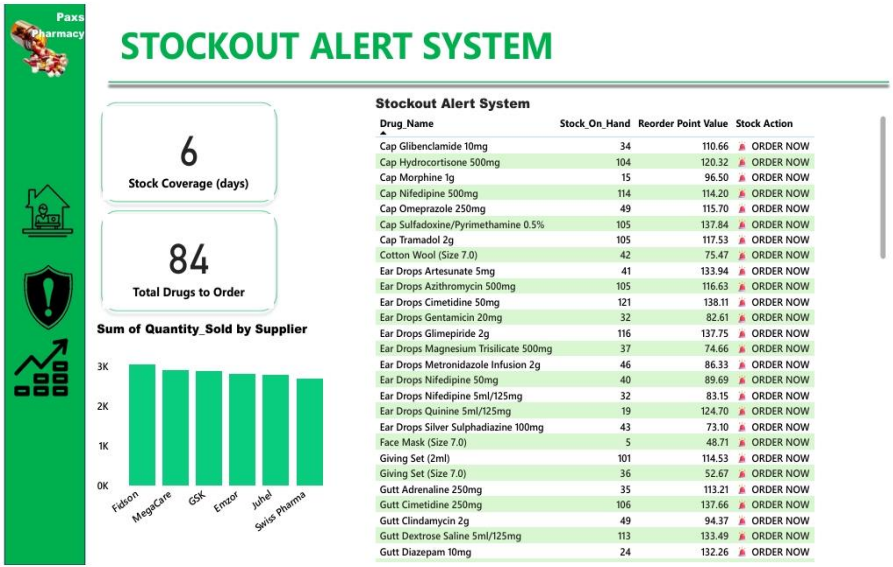
99%
Recall
Expiry Risk

The stockout model correctly identifies 91% out of every 100 at risk drugs. The expiry model is near-perfect — it correctly flags 99% of drugs likely to expire before being sold, almost eliminating blind spots.

POWERBI DASHBOARD VISUALIZATION







Conclusion and Recommendations

The study shows that combining demand forecasting with risk prediction provides a practical system for improving drug inventory management, helping to reduce stockouts and minimize expiry losses.

Recommendations

Pharmacies should use predicted demand together with safety buffers to guide timely restocking decisions

They should also implement continuous monitoring and regularly update the model with new data to maintain accuracy and effectiveness